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Oefeningenreeks 3A nr. 36.20

$$f(x) = \ln\left(\sqrt{\frac{1+x}{1-x}}\right) = \frac{1}{2}\ln\left(\frac{1+x}{1-x}\right)$$

$$f'(x) = \frac{1}{2}\left(\frac{1}{\frac{1+x}{1-x}}\right) \cdot \frac{(1-x)+(1+x)}{(1-x)^2}$$

$$= \frac{1}{2} \cdot \frac{1}{\frac{1+x}{1-x}} \cdot \frac{2}{(1-x)^2}$$

$$= \frac{1-x}{1+x} \cdot \frac{1}{(1-x)^2}$$

$$= \frac{1}{(1+x)(1-x)} = \frac{1}{1-x^2}$$

References